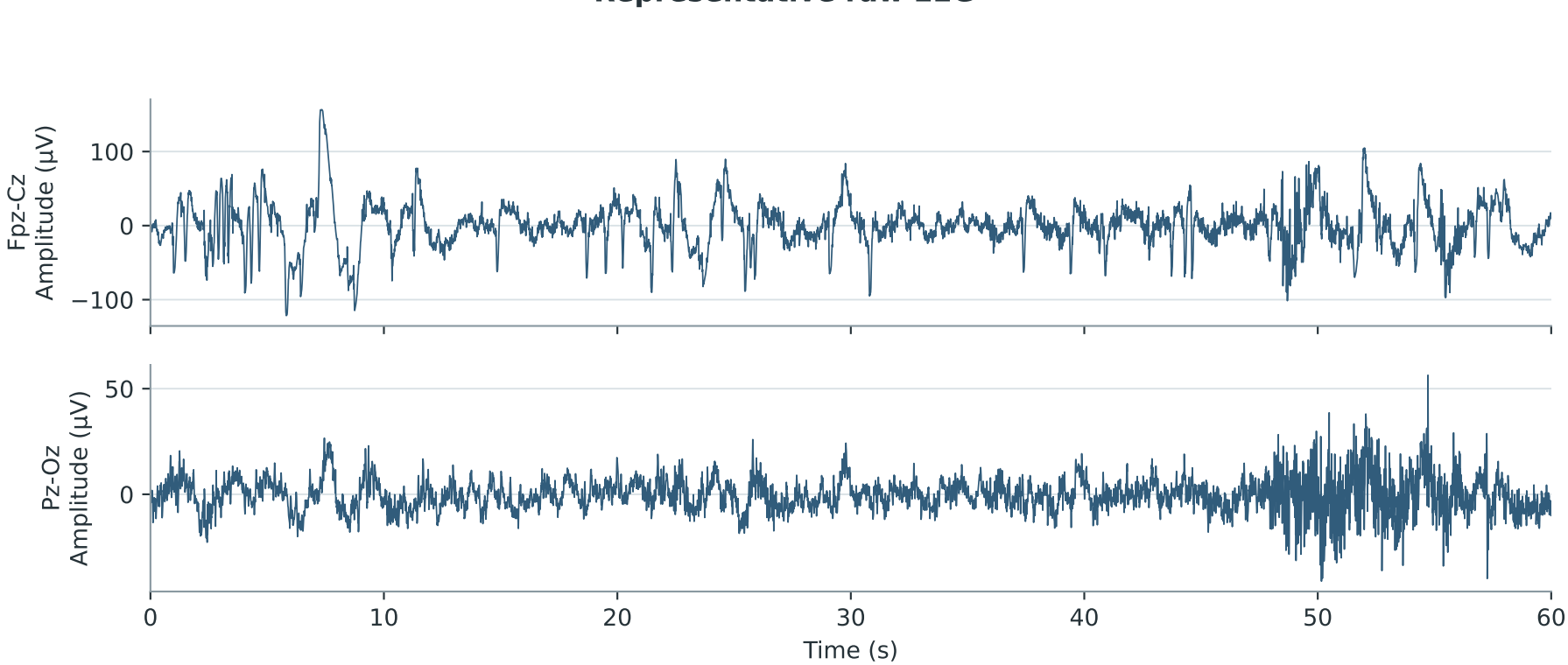
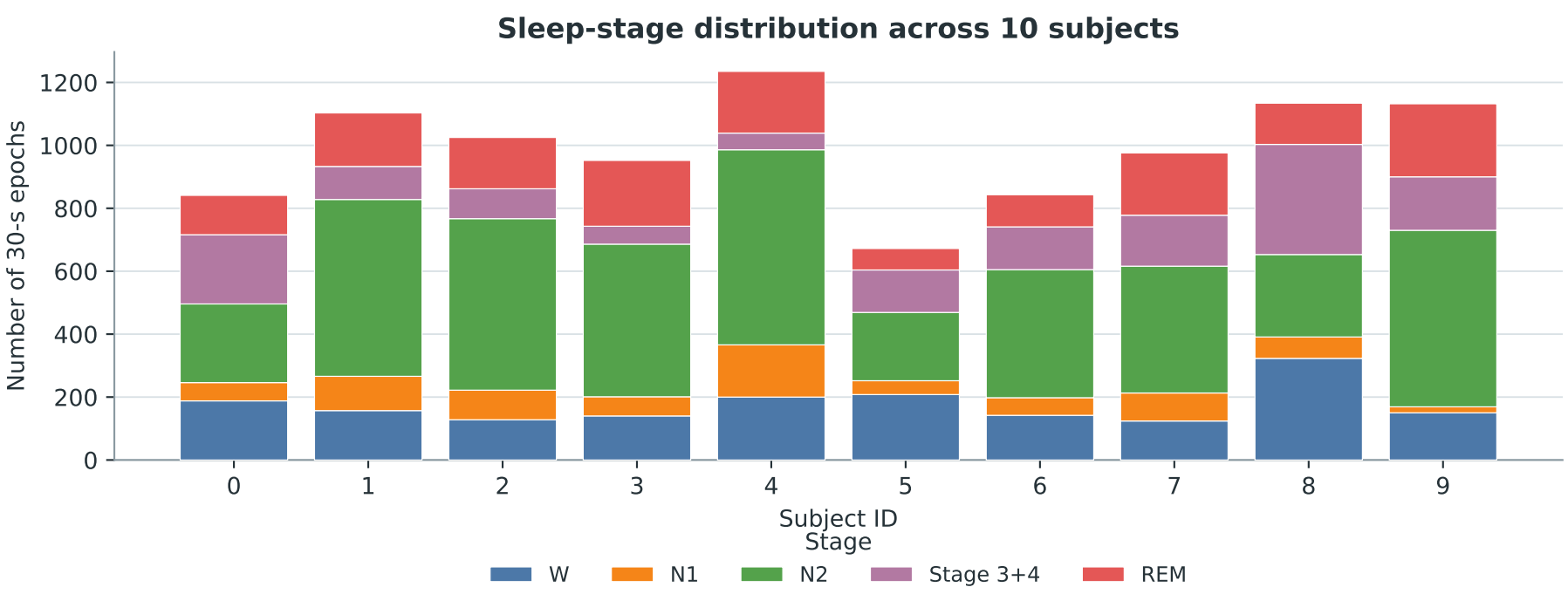
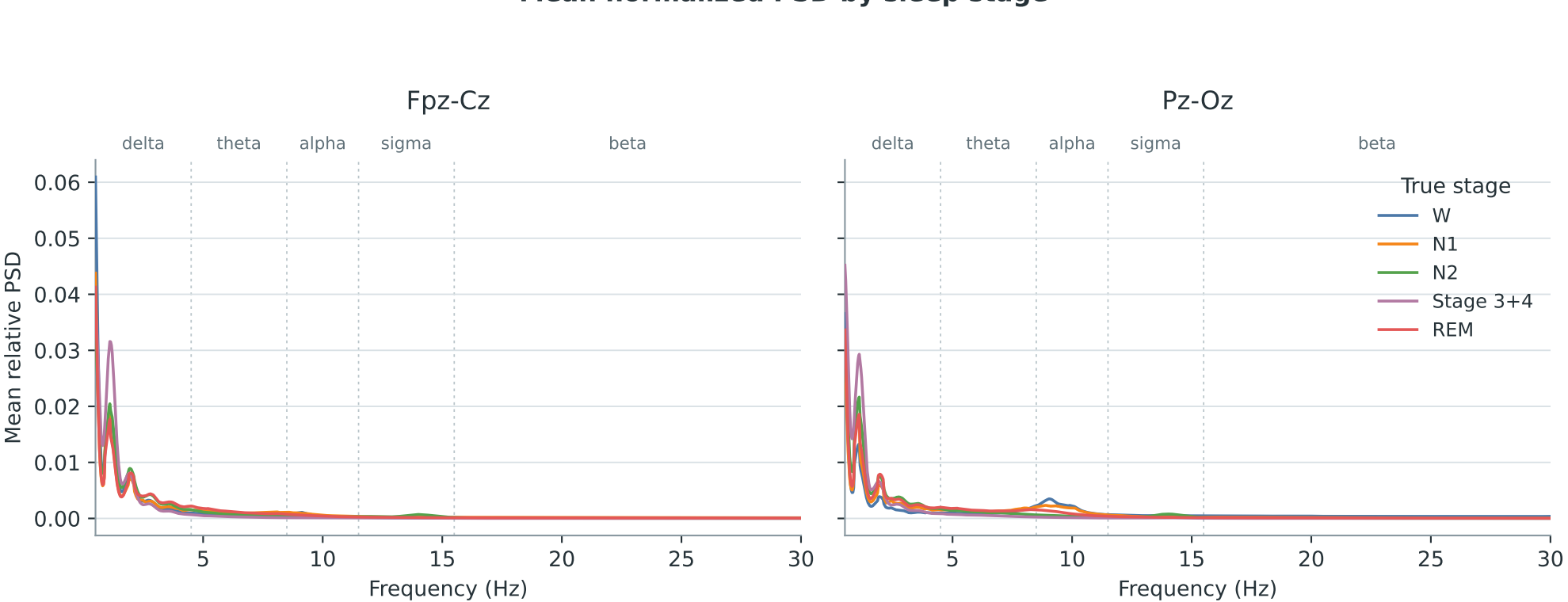
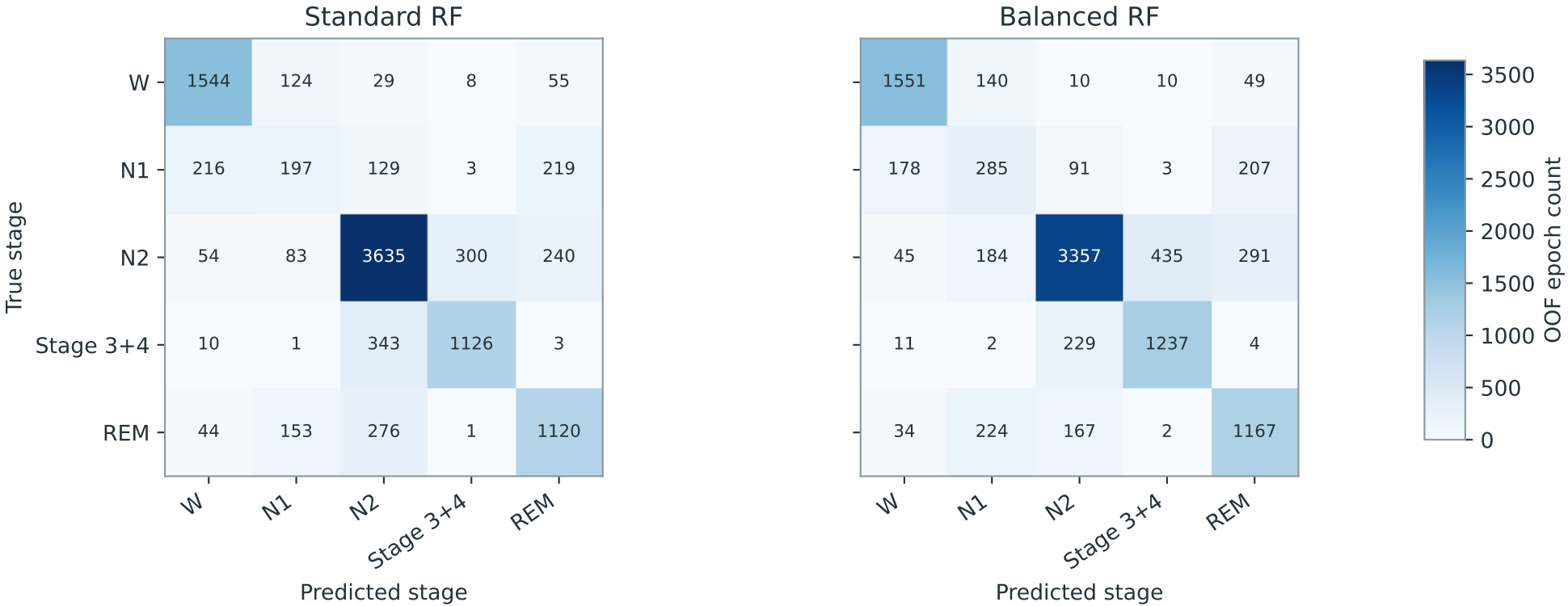


Sleep-EDF subject 0, recording 1; first 60 s; unfiltered Fpz-Cz and Pz-Oz

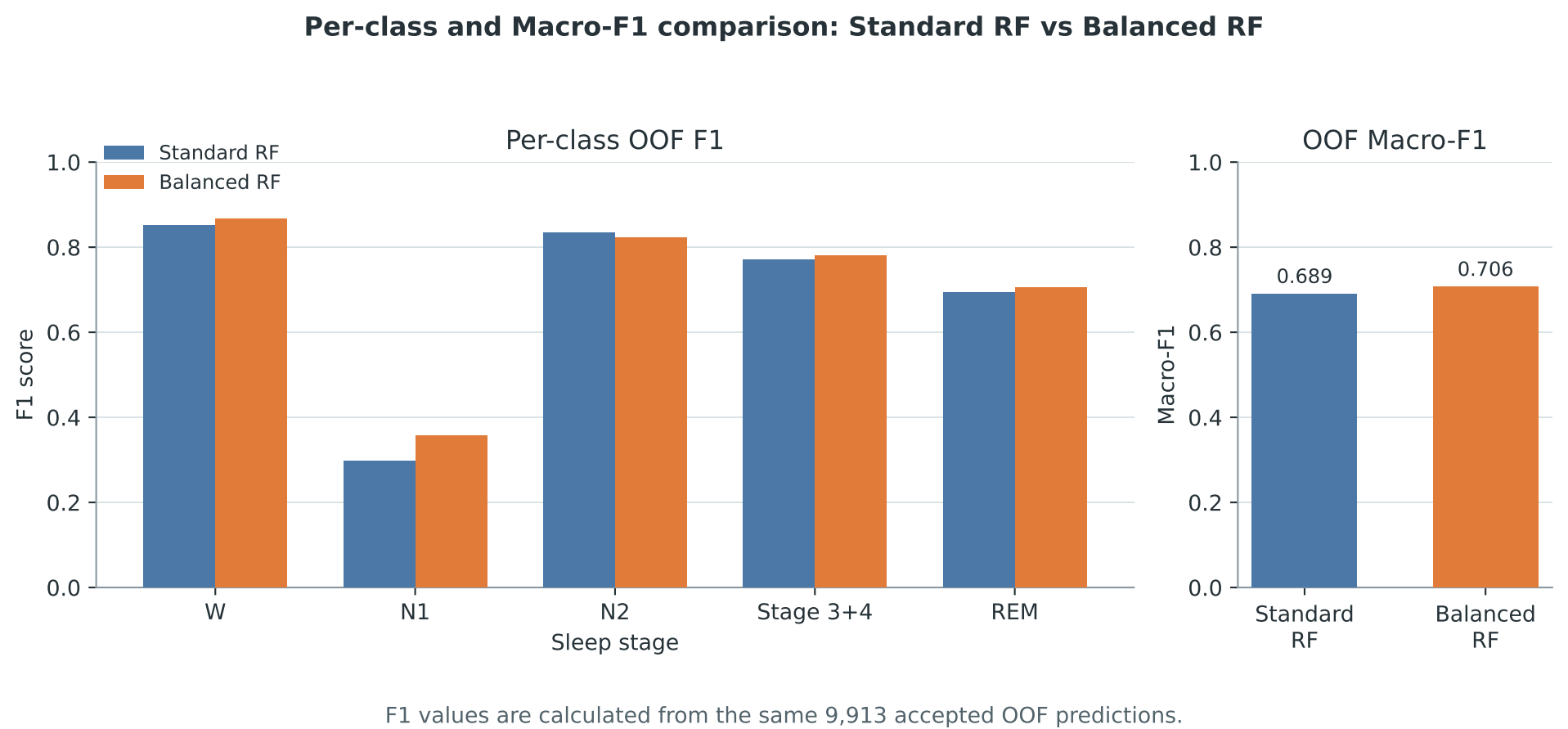


Counts are derived from the accepted OOF truth labels (n = 9,913 epochs).


Mean normalized PSD by sleep stage
Fpz-Cz
Pz-Oz
delta
theta
alpha
sigma
beta
Mean relative PSD
0.06
0.05
0.04
0.03
0.02
0.01
0.00
5
10
15
20
25
30
Frequency (Hz)
True stage
W
N1
N2
Stage 3+4
REM
All 9,913 accepted 30-s epochs; relative PSD normalized within 0.5-30 Hz per epoch and channel.



Rows are true stages and columns are predicted stages; both panels use the same count scale.


Per-class and Macro-F1 comparison: Standard RF vs Balanced RF
Per-class OOF F1
Standard RF
Balanced RF
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
W
N1
N2
Stage 3+4
REM
Sleep stage
OOF Macro-F1
Macro-F1
0.689
0.706
Standard RF
Balanced RF
F1 values are calculated from the same 9,913 accepted OOF predictions.